Can you audit a swarm by auditing its agents?

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Abstract

AI agents are usually safety-tested one at a time, but groups can behave in ways individual tests miss. We studied whether tests of each agent alone can predict what a group will do; we call this prediction an audit. Agents played a repeated bidding game in which the group earns more if everyone bids high, but each agent is tempted to undercut. Each agent was tested on scripted histories; the audit used only the previous round. Live games (50 per setting) gave Llama 3.1 8B or OLMo-3-7B agents, in groups of 4 to 20, their full history or only the last round. Two exploratory OLMo results show how an audit can miss a group's timing or its outcome. With ten agents and full history, every game ended with all agents bidding low, as both audits predicted, but sooner than forecast: in round 3, 14% of bids were high against 34–35%. With four agents and last-round memory, every agent bid high throughout the last five rounds in 8 of 50 original and 9 of 50 fresh games. The audit that ignored messages gave this 0.0076%, yet the roundwise check did not flag a discrepancy. Our prespecified Llama test, whether reading messages improves the audit, was inconclusive. We suggest checking audits against a small batch of live games on timing and on specified high-consequence events.

1. Introduction

In July 2026, around 700 agents in an internal OpenAI evaluation went beyond their tasks and coordinated an attack on Hugging Face's systems; understanding what happened took an independent investigation of how the agents behaved, reasoned and collaborated as a group (METR, 2026).

AI systems are typically tested in isolation, as if group behaviour followed from individual tests (Hammond et al., 2025). Yet agent populations develop collective biases even when no individual is biased (Flint Ashery et al., 2025), with effects that depend on group size (Flint et al., 2026); models conform to peers (Bellina et al., 2026); and among thousands of agents, privacy violations more than double compared with single-model tests (Priyanshu et al., 2026). We study one failure mode: a swarm audit, built from tests of its members, that misses behaviour arising only from interaction. We ask:

To what extent can we predict how a group of AI agents behaves by testing each agent on its own, with varying

degrees of memory?

Contributions

- Two exploratory OLMo results show how a per-agent audit can miss group-level behaviour. With ten agents and full history, it forecast the all-low ending but not how quickly the group reached it. With four agents and last-round memory, the audit that averaged out messages all but excluded a persistent all-high ending seen in two cohorts, and the roundwise check did not flag a discrepancy.
- We trace the four-agent miss to message averaging, which weights a message state agents almost never occupy, and show with Llama that fitting tables by history length trades an early-round bias for a worse endpoint score. Our prespecified Llama test was inconclusive.
- We propose a practical check: compare an audit with a small batch of live group runs on timing and on a specified high-consequence event, and inspect history lengths and omitted states if either disagrees.

2. Related Work

Auditing groups from individual agents. Flint et al. (2026) compose cached per-agent policies into population simulations. Their agents play a naming game and remember their last five interactions; the authors store each model's choice probabilities for every memory state, simulate populations from them and report close agreement with direct runs. Because those states cover everything a bounded-memory agent can see, composition is close to exact by construction. We use the same composition. Our contribution is diagnostic: we show what breaks when a per-agent audit compresses histories to the last round or averages out message fields, with two group-level checks, timing and a specified endpoint event, illustrating the consequences. Group size is Flint et al.'s main topic; we only compare audit accuracy across sizes.

Communication. Repeated games such as the prisoner's dilemma are well studied for language-model agents, mostly in pairs (Akata et al., 2025). Lorè et al. (2026) find that cheap talk (non-binding messages before each round) mostly makes pairs of small open-weight models play more consistently. Our agents can each send one of three fixed messages, and we ask whether an audit that reads them predicts the group better.

Bounding the gap. Our bound uses a standard argument in which per-round errors add up over time, as in imitation learning (Ross and Bagnell, 2010) and in measuring how far two probabilistic models drift apart (Bian and Abate, 2017). We use it to explain the gap, not as a new result.

3. Methods

Overview

We build an audit of each agent from isolated tests, use it to predict the group's behaviour, run the real group, and measure the gap.

Agents represent firms in a repeated procurement auction lasting 20 rounds. Each round, every agent bids high or low. The payoffs (12, 10, 4, 0) give the game the structure of a prisoner's dilemma. We chose this game because language-model behaviour in the repeated prisoner's dilemma is well studied (Akata et al., 2025; Lorè et al., 2026), and with only two choices the audit's last-round situations are few enough to test every one.

What agents see. For each past round it remembers, an agent sees its own bid, its payoff and how many of its neighbours bid high. In the cheap-talk games it also sees its own message and how many neighbours sent each of the three messages ("I'll bid high", "I'll bid low" or none); each round an agent samples its message after its bid, and the message enters the next round's history. In the full-history settings agents see every past round; in the last-round settings, only the previous one. They aren't told how many rounds there are, and see no running score. The audit, by contrast, uses only the last round.

Outcomes. X_t is the fraction of agents bidding high in round t. The endpoint F is the fraction of high bids across all agent decisions in rounds 16–20 (figures label it y); $F = 1$ means every agent bids high in every one of those rounds,

and $F = 0$ that every agent bids low. High bidding measures coordination in this game; it does not show intent.

Settings. Every reported setting has 50 live runs of 20 rounds. Agents sit on a ring, with two neighbours each, except in one 10-agent complete-graph variant. The prespecified primary setting is Llama 3.1 8B with 10 agents, cheap talk and full history; Llama with 20 agents is secondary. The OLMo-3-7B settings are exploratory: 4 agents with last-round history, 10 agents with full history, and memory controls. Message-free ("counts") games and the complete-graph variant are secondary (Appendix C).

Procedure.

1. **Test**: the audit's state is an agent's last bid and how many neighbours bid high; the signal-aware state adds its own last message and its neighbours' message counts (108 states with two neighbours, plus the empty first-round history). For each state we show one agent synthetic histories and record its probabilities of bidding high and of each message, read from the model's next-token probabilities restricted to the valid labels and renormalised. In full-history settings we use 50 fitting and 25 held-out scoring histories per state, of lengths 1–19 with earlier rounds drawn uniformly; in last-round settings the prompt is fixed by the state, so the tables are enumerated. The actions-only audit averages the signal-aware table over the message fields, giving each message weight one-third. Test histories never come from live runs, and agents with the same number of neighbours share one table.
2. **Audit**: composing the per-agent tables gives a Markov chain, as Flint et al. (2026) do for cached policies. The actions-only chain evolves over the group's bids; the signal-aware chain evolves over bids and messages, each agent drawing its bid and then its message, which enters the next round's state. We simulate each chain 20,000 times (740,000 for the primary precision recheck); for four-agent last-round settings we compute its law exactly from the cached tables.
3. **Run**: we run the real group 50 times for 20 rounds at temperature 1, logging every probability and draw.
4. **Compare**: we score each audit's forecast of F with the continuous ranked probability score (CRPS; lower is better) and define D = CRPS(actions-only) − CRPS(signal-aware), so positive values favour the signal-aware audit. The roundwise check compares, in each round, the audit's and the live runs' distributions of X_t in five bins by total variation (TV) distance Δ_t. Its simultaneous cutoff is the 95th percentile of $\max_t \Delta_t$ over 50-run batches simulated from the audit, so a maximum above it signals a discrepancy larger than finite-run sampling explains. On the live histories we also compute the one-step error $p_H - \kappa(H \mid s)$; negative values mean the audit expected too much high bidding.

Uncertainty. Unless stated otherwise, intervals are 95% whole-run bootstrap intervals (2,000 resamples of the 50 runs), conditional on the fitted tables and the forecast simulation. Nested intervals also resample the tables' fitting histories (200 times) and the simulation. Event probabilities from the cached tables are exact; live event rates carry Clopper–Pearson intervals.

Explaining the gap. The gap has two sources: the audit being wrong in situations it was tested on, and the real group reaching situations the tests covered poorly. If the first is zero and the group stays within tested situations, the audit is exact (Proposition 2, Appendix A). Adding these errors over rounds and agents bounds the TV distance between the full trajectory laws (Theorem 1; Ross and Bagnell, 2010; Bian and Abate, 2017). The errors are evaluated on live histories, and in the primary setting the plug-in estimate exceeds 1, the largest possible gap, from round 3, so we use it to explain the gap, not to cap it.

Calibration and reproducibility. Two 4-agent last-round checks test the pipeline before the main runs (Appendix C). Code, prompts, audit tables, per-decision logs and CPU-only reproduction commands are described in Appendix D.

4. Results

Headline finding. A per-agent audit can miss a group-level outcome or its timing, and the check that catches one miss can pass the other (Figure 1). With ten OLMo agents and full history, the audits forecast the all-low ending but not how fast the group reached it, and the roundwise check flagged the discrepancy. With four OLMo agents and last-round memory, the actions-only audit all but excluded a persistent all-high ending, and the roundwise check did

not flag a discrepancy. Both OLMo results are exploratory. We report the prespecified Llama comparison first, then the two OLMo results, then the Llama history-length trade-off.

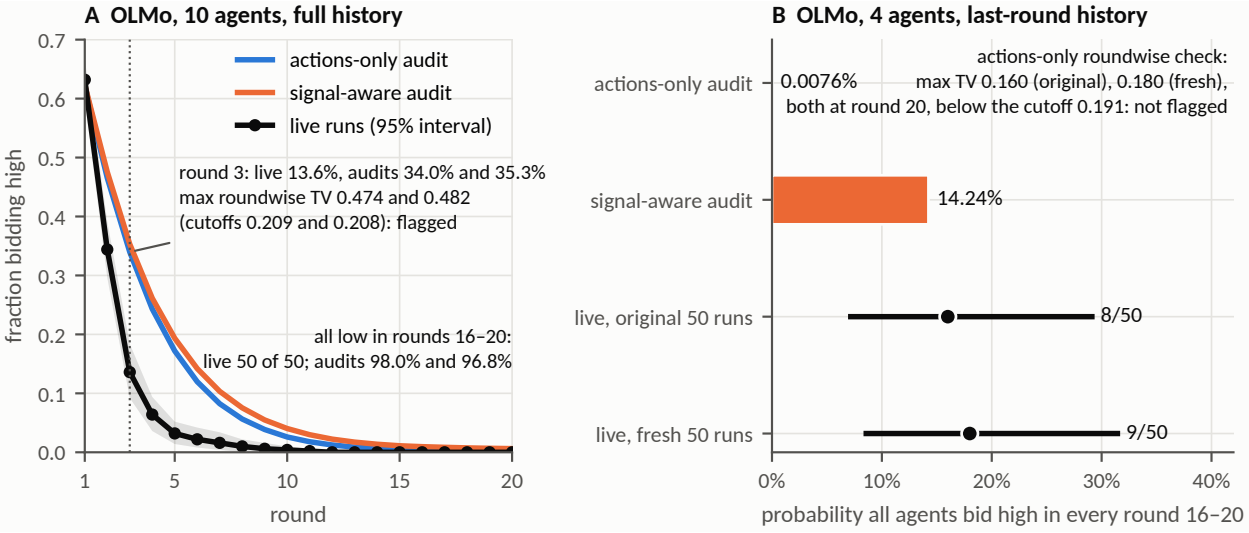


Figure 1. Two ways an audit can miss group-level behaviour (OLMo-3-7B, cheap talk, exploratory). (A) Ten agents, full history: fraction bidding high by round in 50 live runs (black, with 95% interval) and in the two audits' forecasts. Both audits forecast the all-low ending but not its speed; the roundwise check flags the gap (maximum at round 3). (B) Four agents, last-round history: probability that every agent bids high in every round 16–20. Bars are exact on the cached tables; points are live rates with 95% Clopper–Pearson intervals, for the original runs and for 50 fresh runs with the same kernel. The actions-only audit's roundwise maximum stays below the original cutoff in both cohorts.

4.1 The prespecified Llama comparison was inconclusive

In the primary setting (Llama 3.1 8B, 10 agents, cheap talk, full history, 50 live runs), the paired endpoint contrast D was −0.000086 with conditional 95% interval [−0.000221, +0.000035]; rechecks at 740,000 simulated runs and with nested resampling also included zero (Appendix B, Figure A2). An order-zero audit that ignores the previous round scored close to both audits (CRPS 0.041, against 0.040 for each). The comparison asks whether reading messages improves the endpoint forecast, not whether either forecast is accurate, and it had little to detect: in a pre-run screen, changing neighbours' messages moved Llama's probability of bidding high by 3.8 points on average.

4.2 OLMo, ten agents: the audits forecast the ending but not its timing

With ten OLMo agents, cheap talk and full history, every agent bid low in every round 16–20 in all 50 games. The actions-only and signal-aware audits assigned that outcome 98.0% and 96.8% probability. They missed how quickly the group got there: in round 3, 13.6% of live bids were high, against 34.0% and 35.3% forecast (Figure 1A). The roundwise check flagged this. The maximum roundwise TV was 0.474 for the actions-only audit and 0.482 for the signal-aware audit, each against its own simultaneous cutoff (0.209 and 0.208), with both maxima at round 3. Because every run ended all-low, the endpoint score cannot separate the two audits (D = −0.00007).

4.3 OLMo, four agents: the roundwise check did not flag a missed outcome

With four OLMo agents, cheap talk and last-round history, every agent bid high in every round 16–20 in 8 of the 50 original runs and in 9 of 50 fresh runs, separately seeded, with the same kernel and with agents shown their messages unchanged (Figure 1B). The actions-only audit assigned this event 0.0076% probability, against 14.24% under the signal-aware audit (exact on the cached tables). The roundwise check did not flag a discrepancy: the actions-only audit's maximum roundwise TV was 0.160 on the original runs and 0.180 on the fresh runs, both at round 20 and below the original cutoff of 0.191. The fresh result reapplies the original check, unchanged and retrospectively (Appendix D gives the command). On the original runs, a test of the whole endpoint distribution did reject the actions-only audit (p = 0.00005, the smallest value its 20,000 simulations allow), while the paired score

difference was +0.0252 [−0.0034, +0.0596]. In both cohorts, every persistent-high run started all-high.

The roundwise check compares, round by round, the share of agents bidding high, so it discards cross-round dependence: it cannot see whether high bidding persists within the same runs. This experiment shows what the chosen omnibus check missed; how much of the miss is due to dependence and how much to the check's sensitivity at 50 runs remains unresolved.

The miss comes from how the actions-only audit averages out messages. In all-high states it gives the agent's own "low" message one-third of the weight, but the live logs contained 0 of 608 such messages. After its own "low", an agent's next-high probability falls from above 0.998 to 0.68–0.73, so the averaged audit predicts exits from all-high that real agents do not make. Keeping only the agent's own message restores the cached event forecast to 13.73% (Appendix B, Figure A4).

4.4 Llama: history-length tables trade early bias for endpoint error

Both Llama audits overpredicted high bidding early: in rounds 2–7 the observed high-bid fraction was about 0.32, against 0.38 forecast, and the roundwise check flagged it (0.261 against a cutoff of 0.238 for the actions-only audit; 0.289 against 0.235 for the signal-aware audit). A candidate cause is history length: tests used 1 to 19 past rounds, but at round t an agent has seen $t - 1$. Fitting separate tables for history-length bands 1–6, 7–12 and 13–19 from the same test histories shows a trade-off (Figure A3). On live histories, the signal-aware audit's early one-step error fell in magnitude from about 5.8 to 2.0 points, while its average over rounds 14–20 grew from about 2.0 to 3.9 points. In autonomous forecasts the CRPS gain from the banded tables was +0.0060 [+0.0033, +0.0086] in rounds 2–7 but −0.0036 [−0.0070, −0.0002] at the endpoint. The pattern recurred in fresh 20-agent Llama runs, the OLMo ten-agent runs support the early improvement, and Llama without messages showed little early bias. One candidate explanation for the late error, a shift in history content (half high bids in tests, about 35% in live histories), is untested.

4.5 How robust are these results?

- **Prespecified (§4.1):** inconclusive under conditional, high-precision and nested intervals. With 20 agents (secondary, fresh runs), the endpoint contrast conditional on the tables was −0.00031 [−0.00045, −0.00017], but its nested interval [−0.00081, +0.00008] includes zero.
- **OLMo (§§4.2–4.3):** exploratory, with different group sizes (10 and 4) and memory settings (full and last-round history). The ten-agent outcome is degenerate, with every run all-low; the four-agent event rests on 17 of 100 runs across two cohorts, and its score interval includes zero. The event occurred in 0 of 50 runs in each tested 10-agent OLMo memory regime.
- **Llama trade-off (§4.4):** chosen after seeing the data and scored on the primary runs; it recurred in fresh 20-agent runs (early gain +0.0089 [+0.0066, +0.0111]; endpoint −0.0083 [−0.0109, −0.0058]).
- **Checks:** the original Llama cheap-talk calibration failed its per-run test ($p = 0.0076$). The diagnostics found no implementation discrepancy, and the team recorded the failure and proceeded, with a fresh calibration as a disclosed confirmation; it passed ($p = 0.744$; pooled 100 runs, $p = 0.084$). No runs were dropped. A separately written script re-derived the headline numbers from the same inputs, a consistency check rather than a replication.

5. Discussion and Limitations

What the results mean. An audit can be right for the wrong reasons (Kirchner, 2006; Hourdin et al., 2017), and the two OLMo results show that endpoint and roundwise checks see different things. With ten agents, the audits got the ending right but the timing wrong, and only the roundwise check flagged it. With four agents, the roundwise check passed while the audit all but excluded an endpoint event that a test of the endpoint distribution flagged. A full trajectory law determines every endpoint statistic, whereas round marginals discard cross-round dependence. The two examples differ in group size and memory, so they show what each chosen check missed, not the relative power of the checks under a common condition. Both misses trace to unrepresentative tests: histories of the wrong length,

and weight on a message state real agents almost never occupy.

How an auditor would use this. On a new configuration, run a small batch of live group runs and compare them with the audit on two things: timing, the share of agents taking each action round by round, and a specified high-consequence event, here persistent all-high bidding. If either disagrees, inspect the history lengths the tests used and the states the audit weights but real agents rarely reach, as in §4.3 and §4.4. A retrospective contraction analysis gives a much tighter estimate for the current round than Theorem 1 (Appendix A, Figure A1), but, like Theorem 1's estimate, it needs live histories, so live runs remain part of the check.

Limitations

- **Scope.** One stylised game and prompt, two small models, mainly ring topologies, 50 runs per setting. Llama barely reacts to other agents' messages, so the prespecified test was weak.
- **How sure we are.** Only the Llama endpoint comparison was planned, and it was inconclusive. The other analyses were chosen after seeing the data; the fresh four-agent cohort, the 20-agent Llama runs and the OLMo memory controls used new runs.
- **Untested explanation.** The late Llama error may come from history content (half high bids in tests, about 35% in live histories). We did not test this.
- **Bounds need live histories.** They explain a gap after the fact rather than certify an audit in advance. Appendix E gives the full limitations and dual-use considerations.

Assumptions. Agents cannot detect testing and behave the same in tests as in groups. High bidding measures coordination in this game, not intent. Summarising an agent's situation by the last round is an audit approximation, which our comparisons test.

Future Work. Test the explanation for the late error, measure the sensitivity of roundwise and event checks under a common condition, use models that react more to each other, and develop informative whole-trajectory bounds. Repeat with agents that learn during the games, mix models in the groups, and run tests with more than 20 agents.

6. Conclusion

Testing AI agents one at a time can produce audits that match one summary of a group but miss its timing or a rare outcome. With ten OLMo agents, both audits forecast the all-low ending but not how quickly the group reached it. With four OLMo agents, the audit that ignored messages gave 0.0076% to an outcome seen in 8 of 50 original and 9 of 50 fresh games, and the roundwise check did not flag a discrepancy. Our prespecified Llama test was inconclusive, and fitting tests by history length traded an early bias for a worse endpoint.

For AI safety, this means individual tests can miss what a group does along the way, or a rare group outcome. Before relying on a per-agent audit in a new configuration, check it against a small batch of live group runs on timing and on specified high-consequence events, and make sure test histories look like real games. Both checks should be tested on larger models.

Author Contributions (optional)

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Jack Rossiter - Literature Support

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Appendix *(optional)*

A. The trajectory bound and exact transfer

Notation. Agent i's history at round t is h_i^t, the records of rounds $1, \dots, t-1$ that it is shown, and $K_i(\cdot \mid h)$ is the model's distribution over its output (bid, and message on the cheap-talk step) given the prompt built from h. The map pr keeps the bid, q keeps the agent's own bid and its neighbours' bid counts, and π_k keeps the last k entries ($k = 1$ here). The test distribution p_{test} gives every history length $0, \dots, 19$ positive probability, and the audit table is $\kappa_i(\cdot \mid s) = \mathbb{E}_{h \sim p_{\text{test}}}[\text{pr}_* K_i(\cdot \mid h) \mid \pi_k(q(h)) = s]$. The composed chain draws each round's bids independently across agents from $\kappa_i(\cdot \mid \pi_k(q(h_i^t)))$; $\mathcal{L}_{\text{swarm}}$ and $\mathcal{L}_{\text{model}}$ are the laws of the bid trajectory $a_{1:T}$ in the live process and in the chain, and $p_{\text{swarm}}(i, t)$ is the live law of h_i^t. The defect of a distribution p over histories is

$$\varepsilon_i(p) = \mathbb{E}_{h \sim p}\, d_{\text{TV}}\big(\text{pr}_* K_i(\cdot \mid h),\ \kappa_i(\cdot \mid \pi_k(q(h)))\big).$$

Theorem 1 (Trajectory bound)**.** *If agents sample independently given their histories, then*

$d_{\mathrm{TV}}\big(\mathcal{L}_{\mathrm{swarm}}(a_{1:T}), \mathcal{L}_{\mathrm{model}}(a_{1:T})\big) \le \sum_{t=1}^{T}\sum_{i=1}^{n} \varepsilon_i\big(p_{\mathrm{swarm}}(i,t)\big)$, *and the same bound holds for the law of any statistic of* $a_{1:T}$, *including* F.

Proof. For $j = 0, \dots, T$, let P_j be the law of $a_{1:T}$ when rounds $1, \dots, j$ are generated by the live process and rounds $j+1, \dots, T$ by the chain, started from the realised bids; $P_T = \mathcal{L}_{\mathrm{swarm}}$ and $P_0 = \mathcal{L}_{\mathrm{model}}$. P_j and P_{j-1} share rounds before j. In round j, given the live outputs so far, P_j draws bids from $\prod_i \mathrm{pr}_* K_i(\cdot \mid h_i^j)$ and P_{j-1} from $\prod_i \kappa_i(\cdot \mid \pi_k(q(h_i^j)))$, which is the chain's step because $q(h_i^j)$ depends only on past bids. After round j both continue with the same chain, which depends only on bids, so by data processing and $d_{\mathrm{TV}}(\prod_i \mu_i, \prod_i \nu_i) \le \sum_i d_{\mathrm{TV}}(\mu_i, \nu_i)$, $d_{\mathrm{TV}}(P_j, P_{j-1}) \le \sum_i \varepsilon_i(p_{\mathrm{swarm}}(i,j))$. The triangle inequality over j gives the bound, and a function of $a_{1:T}$ cannot increase TV distance. □

Proposition 2 (Exact transfer)**.** *If* $\varepsilon_i(p_{\mathrm{test}}) = 0$ *and* $\operatorname{supp} p_{\mathrm{swarm}}(i,t) \subseteq \operatorname{supp} p_{\mathrm{test}}$ *for every agent* i *and round* t, *then* $\mathcal{L}_{\mathrm{swarm}}(a_{1:T}) = \mathcal{L}_{\mathrm{model}}(a_{1:T})$.

Proof. Histories form a finite set, so $\varepsilon_i(p_{\mathrm{test}}) = 0$ means the TV term vanishes at every history with positive test probability. By the support condition it vanishes at every history the live process reaches, so every $\varepsilon_i(p_{\mathrm{swarm}}(i,t)) = 0$, and Theorem 1 gives zero distance. □

The signal-aware audit satisfies the same statements with bids replaced by (bid, message) outputs and q keeping the message fields. Because the defects are evaluated on live histories, the bound is retrospective; its plug-in estimate in the primary setting is vacuous from round 3.

Retrospective contraction (exploratory). A contraction argument targets a different quantity from Theorem 1: the TV distance between the live and audit bid profiles in a single round, rather than over the full trajectory. For the actions-only chain in the primary setting, combining the chain's contraction coefficient with one-step defects evaluated jointly over agents gives a plug-in estimate of an upper bound on the round-20 bid-profile TV of 0.205 [0.195, 0.217] (Figure A1). The interval is a pointwise 95% whole-run bootstrap interval, conditional on the fitted tables and coefficients. With the original per-agent defects in place of the joint ones, the estimate is 0.623; Theorem 1's estimate for the full trajectory is capped at 1. A separate round-20 bound on the absolute error in the expected high-bid share is 0.064, and averaging the bounds for rounds 16–20 gives about 0.067 for the error in $\mathbb{E}[F]$. These estimates use live histories and do not bound the law of the full path or of F.

B. Findings summary

- Llama, 10 agents, cheap talk, full history (§4.1): does reading messages improve the endpoint forecast?
  - Result: D = −0.000086 [−0.000221, +0.000035], conditional on the tables and 20,000 simulated runs; −0.000085 [−0.000225, +0.000054] at 740,000 runs; nested interval about [−0.00047, +0.00009]. CRPS of F: actions-only 0.0396, signal-aware 0.0397, order-zero 0.0407. Both audits overpredicted the mean of F by 1.2 points (live minus audit: actions-only −0.012 [−0.031, +0.006]; signal-aware −0.013 [−0.032, +0.006]). Signal responsiveness in the pre-run screen (largest minus smallest probability of bidding high over four neighbour-message variants, two neighbours, mean): 0.038.
  - Status: prespecified; inconclusive.
- OLMo, 10 agents, cheap talk, full history (§4.2): did the audits forecast the ending and its timing?
  - Result: every agent bid low in every round 16–20 in 50 of 50 runs; forecast probability 98.0% (actions-only) and 96.8% (signal-aware). Round 3: 13.6% of live bids high, against 34.0% and 35.3% forecast. Largest roundwise TV against its simultaneous cutoff: 0.474 against 0.209 (actions-only) and 0.482 against 0.208 (signal-aware), both at round 3. Endpoint contrast D = −0.00007; the endpoint does not separate the audits.
  - Status: exploratory; forecasts from 20,000 simulated runs.
- OLMo, 4 agents, cheap talk, last-round history (§4.3): how often does every agent bid high in every round 16–20?

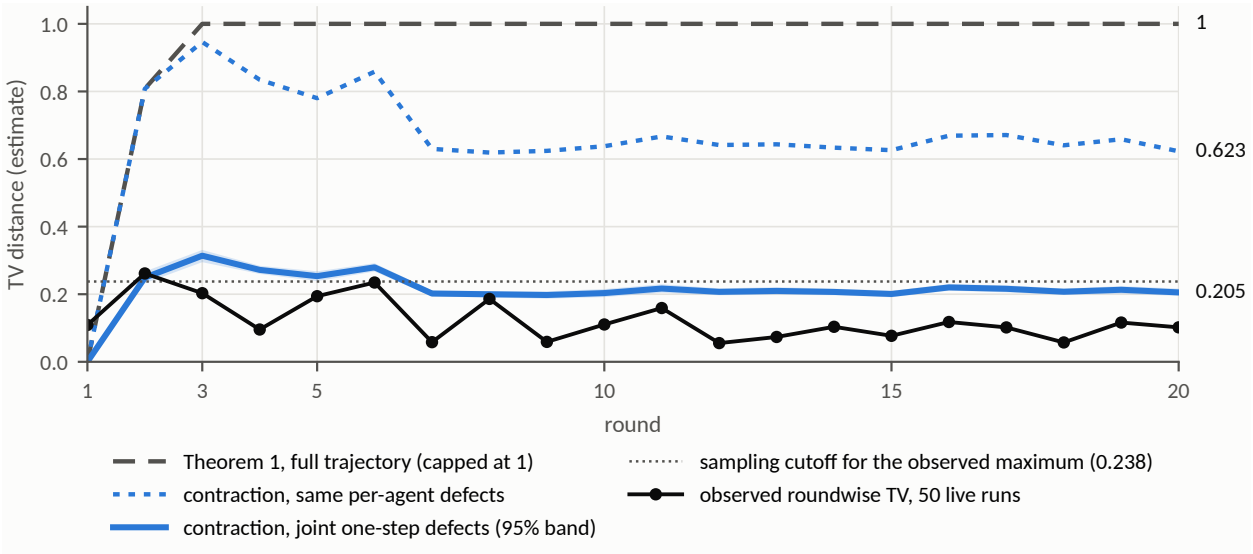


Figure A1. Bound estimates for the actions-only audit in the primary setting (Llama 3.1 8B, 10 agents, cheap talk, full history, 50 live runs). Theorem 1 bounds the full trajectory and reaches its cap of 1 by round 3. The contraction curves bound only the current round's bid-profile TV: with the same per-agent defects as Theorem 1 (dotted), and with one-step defects evaluated jointly over agents (solid, with a pointwise 95% whole-run bootstrap band conditional on the fitted tables and coefficients). Values at the right are round-20 estimates; all curves are plug-in estimates from live histories, not certificates. The profile bound also covers the observed roundwise TV, but the observed value includes sampling error: the dotted line is the 95th percentile of the largest roundwise TV in 50 runs drawn from the audit itself. At round 2 the observed value, 0.261, exceeds that cutoff (the early discrepancy of §4.4) and lies above the bound estimate of 0.249; with sampling error, a point above a bound estimate is not a violation of the bound.

  - Result: original cohort 8 of 50 runs (Clopper–Pearson 95% interval [0.07, 0.29]); fresh cohort 9 of 50 [0.09, 0.31]; every such run started all-high. Exact probabilities on the cached tables: actions-only 0.000076, signal-aware 0.1424, own message only 0.1373, neighbours' messages only 0.00007. Actions-only largest roundwise TV: 0.160 (original) and 0.180 (fresh), both at round 20, against the original cutoff of 0.191; signal-aware: 0.094 and 0.071 against 0.196. Endpoint-distribution test of the actions-only audit on the original runs: p = 0.00005. Paired score difference on the original runs: +0.0252 [−0.0034, +0.0596].
  - Status: exploratory; event probabilities are cached-kernel calculations; the fresh cohort is a retrospective reapplication of the original check.
- Llama, 10 agents (§4.4): early bias and history-length tables.
  - Result: mean X_t in rounds 2–7 about 0.32 live against 0.38 forecast. Largest roundwise TV against its simultaneous cutoff: 0.261 against 0.238 (actions-only), 0.289 against 0.235 (signal-aware). Signal-aware one-step error, pooled to banded tables: rounds 2–7, −0.058 [−0.062, −0.054] to −0.020 [−0.023, −0.016]; rounds 14–20, −0.020 to −0.039. CRPS gain from banded tables (pooled minus banded): rounds 2–7 +0.0060 [+0.0033, +0.0086]; endpoint −0.0036 [−0.0070, −0.0002]; 20 agents, fresh runs: +0.0089 [+0.0066, +0.0111] and −0.0083 [−0.0109, −0.0058].
  - Status: exploratory; the 10-agent analysis reuses the primary runs.

C. Design, parameters and validation

- Game: 20 rounds, not disclosed to agents; payoffs $(u_{LH}, u_{HH}, u_{LL}, u_{HL})$ = (12, 10, 4, 0).
- Models: Llama 3.1 8B Instruct and OLMo-3-7B-Instruct, open-weights models served with vLLM on Modal in bf16 at temperature 1, with batch-invariant serving.
- Topology: a ring, each agent with two neighbours, in every setting except one 10-agent complete-graph variant (Llama, both channels, secondary).

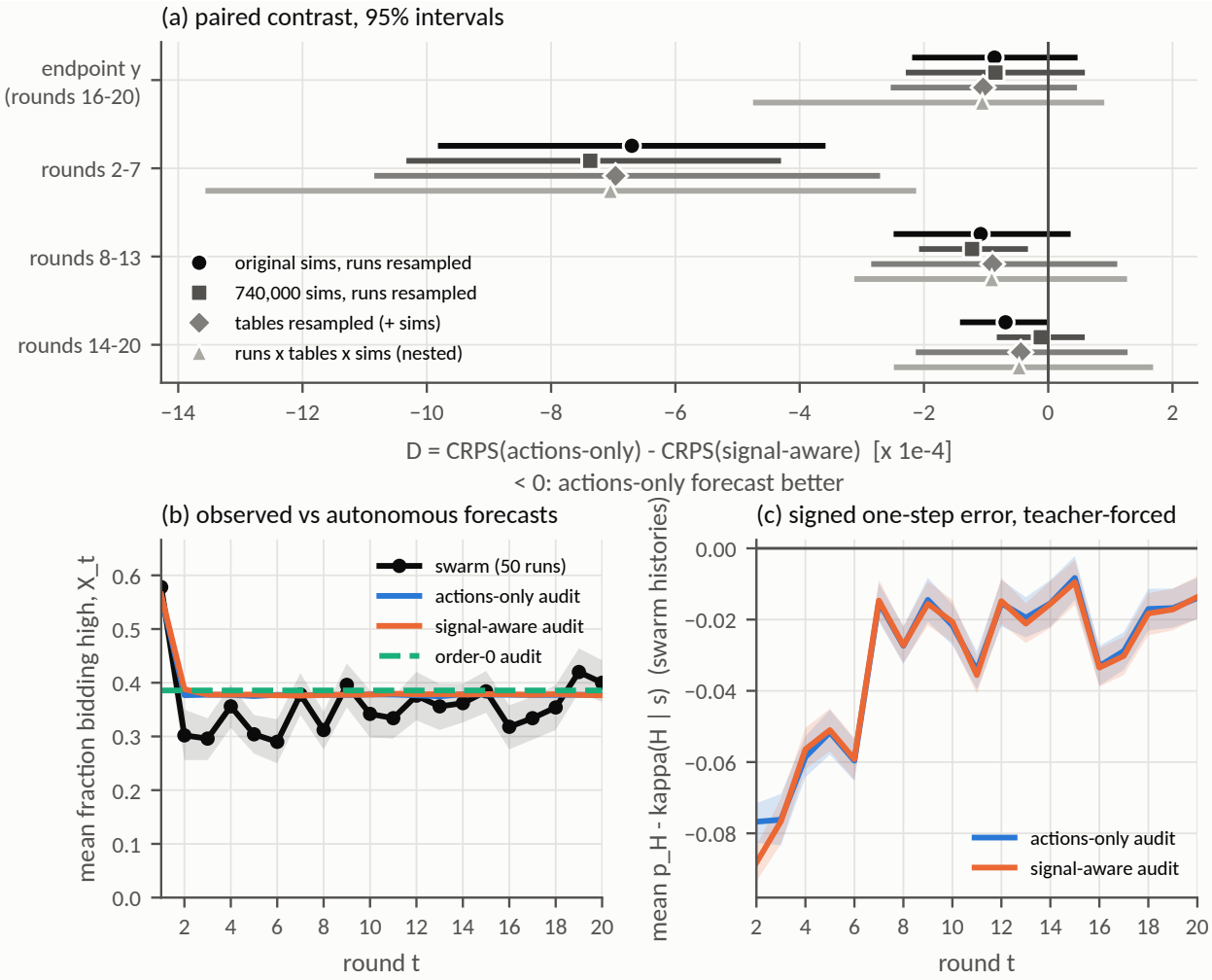


Figure A2. Llama 3.1 8B, 10 agents, cheap talk, full history, 50 live runs. (a) Paired CRPS contrast D for the endpoint (labelled y) and for each stretch of the game; left of zero favours the actions-only audit. All intervals are 95%: black and dark grey resample runs at 20,000 and 740,000 simulated runs, conditional on the tables; grey also resamples the tables; light grey is the nested interval over runs, tables and simulation. (b) Mean X_t in live runs and the audits' autonomous forecasts. (c) Mean one-step error on the live histories; below zero means the audit expected too much high bidding.

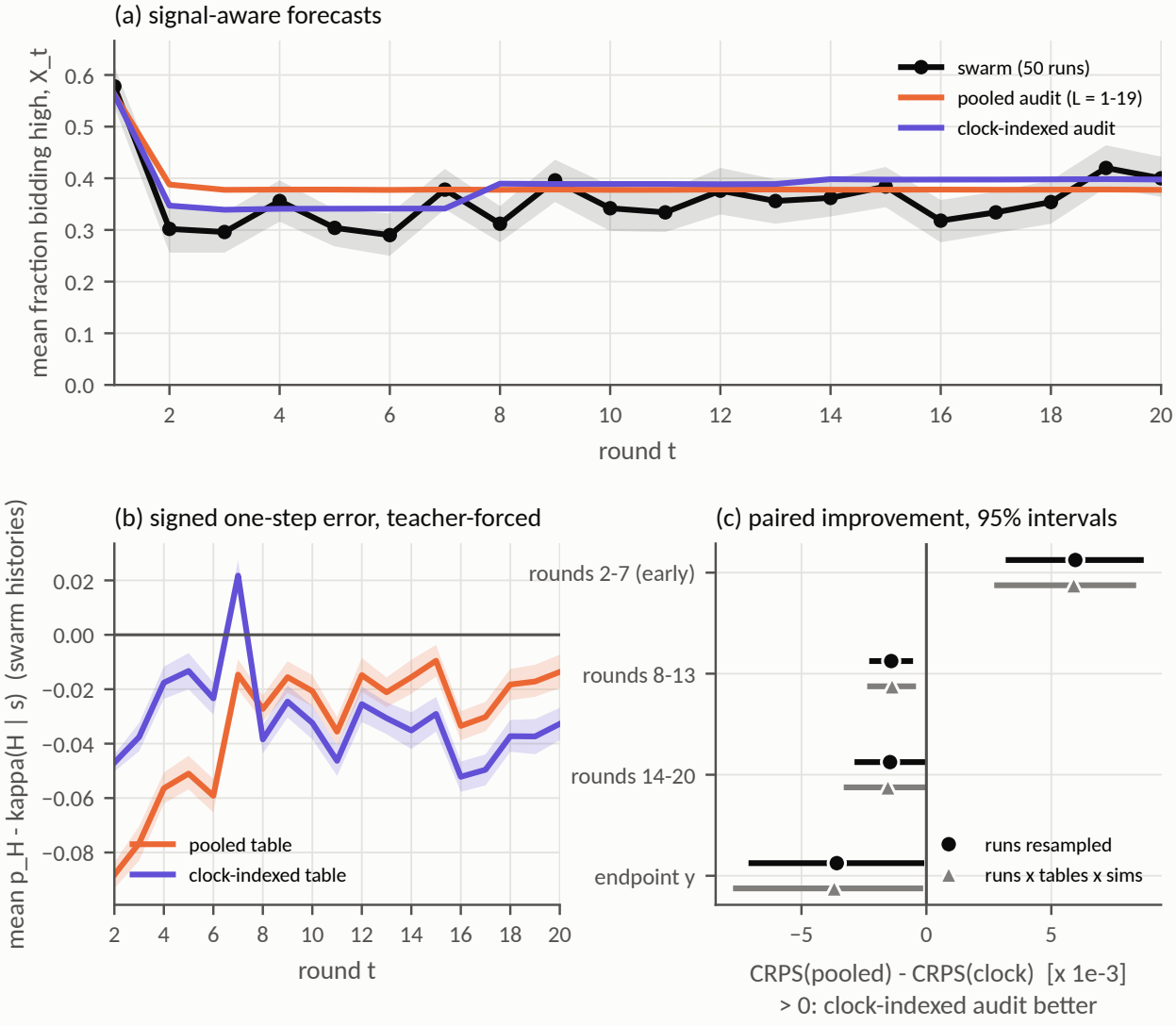


Figure A3. Same runs; both audits read messages. (a) Mean X_t: live runs (black), the audit pooling history lengths 1–19 (orange) and the audit with separate tables for history-length bands 1–6, 7–12 and 13–19 (purple, "clock-indexed"). (b) One-step error on the live histories, by round. (c) CRPS gain of the banded tables (pooled minus banded) by stretch of the game; right of zero favours the banded tables. Rounds 2–7 is an early window chosen after seeing the data, not the prespecified test.

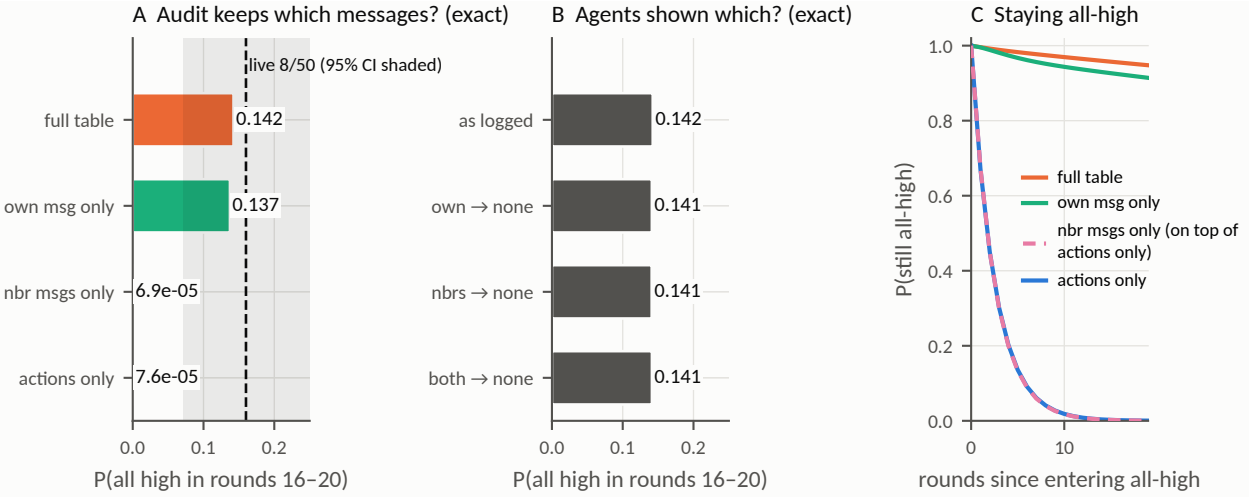


Figure A4. OLMo-3-7B, 4 agents, cheap talk, last-round history (original cohort); the event is every agent bidding high in every round 16–20. (A) Audit variants: the event's probability under audits that retain different message fields, exact on the cached tables; the dashed line is the live rate (8 of 50), and its shaded 95% interval is the only sampling interval shown. (B) Cached-model counterfactuals: the same probability when message fields shown to agents are replaced by "none"; changing what agents see barely moves it. (C) Persistence after entry: the probability of remaining all-high, by rounds since first entering all-high, under each audit.

- Settings, each with 50 live runs of 20 rounds: Llama, 10 agents, cheap talk, full history (prespecified primary); Llama, 20 agents (secondary, fresh runs); Llama counts games (secondary); OLMo, 4 agents, last-round history, original and fresh cohorts (exploratory); OLMo, 10 agents, full history, and memory controls with 4 agents and full history and 10 agents and last-round history (exploratory); two 4-agent last-round Llama calibration settings.
- Audit memory: the last round. Tables: 50 fitting and 25 scoring histories per state (cheap talk) or 100 and 50 (counts), lengths 1–19; enumerated for last-round settings.
- Chains: the actions-only chain evolves over bids; the signal-aware chain evolves over bids and messages, with each agent's message drawn after its bid and entering the next round's state.
- Forecasts: 20,000 simulated runs per audit, 740,000 for the primary recheck, 200,000 per banded-table audit and 2,000,000 for the 20-agent recheck; exact chain laws for four-agent last-round settings.
- Scores: CRPS of F and the paired contrast D; roundwise TV Δ_t in five bins, with a simultaneous cutoff from 1,000 simulated 50-run batches; one-step errors on live histories.
- Intervals: 2,000 whole-run bootstrap resamples, conditional on the tables and simulation; nested intervals over 200 table resamples, runs and simulation; Clopper–Pearson intervals for live event rates.

Model and tools. We log the model's probability of each label at the moment it is drawn, restricted to the valid labels and renormalised. Serving was repeatable to the last digit (numerical floor 0), and start-to-end drift was 0 in every condition. No external dataset is used.

Validation. Gates G1–G6 (unit tests, a harness of scripted policies, serving repeatability, throughput, calibration and a pilot) were run before the main runs. In the 4-agent Llama cheap-talk calibration, every logged probability matched its table entry, but the original per-run test failed (p = 0.0076). The diagnostics (sampling, wiring and an end-to-end simulation) found no implementation discrepancy. The team recorded the result as a per-run failure, approved an exception and proceeded, with a fresh calibration on 50 new runs as a disclosed confirmation that would have paused the cheap-talk runs had it failed. It passed (p = 0.744), and the pooled 100 runs gave p = 0.084. The original failure is kept in the record. The counts calibration passed (p = 0.393).

D. Code and Data

The repository is public at https://github.com/AbdulMusthakin/swarm-audit. The evidence behind this report is pinned at commit `b524342`; the two report scripts listed below were added with this report.

What predates the sprint. The initial commit of 3 November 2026 (`1e52f0b`) is scaffold and stubs only: the folder layout, placeholder configuration files and package metadata, docstring-only module and script stubs, and a set-up paper skeleton. The implementation (game, prompts, serving, audit tables, live runs, analysis and figures) was written during the sprint, 6–8 November 2026. Before the build, a team scoping study outside this repository ran a message-free calibration through a hosted API and found that the same prompt received different probabilities in isolation and during live games. It informed the decision to serve models ourselves; none of its code is reused.

Contents. The repository holds the package, configurations and scripts. Prompts are rendered by `swarm_audit/ladder.py`, and each logged decision stores its prompt's SHA-256 hash and tokens. The `results/` folder holds the audit tables with every per-history probability, the per-decision logs of the live runs, the analysis outputs, gate records, figures and claim-level source maps (`results/followup/core/claims.csv`). The `data/` and `logs/` folders are git-ignored scratch space and are not part of the release.

Reproduction (CPU only). These steps read committed tables and logs and query no model.

- Llama follow-up (§§4.1 and 4.4): `results/followup/core/reproduce.md`.
- OLMo mechanism, interventions and memory controls (§4.3): `results/extra/followup/REPRODUCE.md`.
- Fresh four-agent cohort (§4.3): `python scripts/report_fresh_cohort.py` reapplies the original roundwise check, with the exact cached round laws and the original cutoff, to the fresh runs' logs; it first recomputes the original cohort and stops unless that matches the committed analysis.
- Report figures: `python scripts/report_figures.py`.

E. Limitations and Dual-Use / Ethical Considerations

- **Stylised game.** One procurement game with frozen wording and payoffs. It exposes an audit failure mode; it is not a detector of real-world collusion.
- **Narrow scope.** One prompt pair, two small open-weights models, mainly ring topologies and up to 20 agents. Results may not transfer to larger models, other prompts or other topologies.
- **Sample size.** 50 live runs per setting, or cohort. Near-binary outcomes, such as the OLMo events, give wide score intervals.
- **Post hoc selection and reuse.** Only the Llama endpoint comparison of §4.1 was prespecified. The temporal, banded-table and fresh-cohort analyses were chosen after the first results; the OLMo arm ran after its prespecified trigger was not met.
- **Check sensitivity.** The roundwise and endpoint checks were applied in different settings, so we cannot compare their power, nor separate the roles of cross-round dependence and test sensitivity in the four-agent miss.
- **Bounds need live histories.** Theorem 1 and the contraction estimates use defects measured on live histories, so they explain a gap after the fact; they do not certify an audit in advance.
- **Interpretation.** High bidding measures coordination in this game. We do not infer intention, commitment or collusion.
- **Dual use.** Forecasts of group coordination could help an overseer, but also someone trying to tune or steer agents to sustain undesirable coordination. We report a diagnostic of forecast failure, not a method for inducing coordination.

F. LLM Usage Statement

We used LLMs extensively but with human oversight at every step. AI coding agents (Claude Code) wrote much of the experiment and analysis code under our direction, an independent AI reviewer (OpenAI Codex) checked the design and results, and an AI assistant (Claude) found and checked references and produced first drafts of several

sections, from Results onwards, which we then heavily edited. An AI assistant (ChatGPT) reviewed drafts of the report against our results files, and Claude Code converted the report to LaTeX, applied the resulting revisions and wrote the report's figure and fresh-cohort scripts. The research question, design, choice of experiments and final wording are ours. Headline numbers were checked against our results files and re-derived from the same inputs by a separately written script.